Semaev vs hashed relations as constraint graphs: a treewidth census on toy curves

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Labels. **[measured]** read from a recorded row (`rows/*.json`); **[exact]** recomputed by the code from the object itself (a width from its executed order, an acceptance from the network); **[heuristic]** an upper bound from a greedy order; **[bound]** a lower bound with its argument; **[derived]** arithmetic on measured values, shown. The layout (`render.py`) computes nothing about any network.

1. Question and instrument

The desk audit `research/hashed_ecdl_tensor_audit_20261107` closed the Bochum hashed-factor-base contraction lead on admission and constants: the generic contraction theorem (openai/math Theorem D.1) never engages below 2^{131071} edges, and a network of s edges is charged $2^{s/\sqrt{\log s}}$ with no credit for structure. The owner's follow-up: Semaev summation polynomials are genuine algebraic relations; encoded as constraint graphs they might give a solver structure to exploit, where hashed relations carry none. The first experiment: build the $O(b)$-edge point-addition graph, measure treewidth and contraction cost on toy curves (b about 20–40), and compare against the bound, to see whether the encoding or the solver is the bottleneck.

Treewidth is the right instrument. A Boolean tensor network whose primal graph (one variable per edge, one clique per vertex) has treewidth w is contracted exactly by variable elimination in $O(V \cdot 2^{w+1})$ table operations whatever s is; up to the choice of order this is the best any structure-exploiting exact contraction can do, and it is exactly what the D.1 bound ignores. So for each network we record a lower bound on w (what no order can beat) and the best upper bound several orders achieve, and set both against the three reference exponents: the D.1 bound $s/\sqrt{\log_2 s}$ at constant 1, naive enumeration of the free slots $(m-1)b/m$, and rho $b/2$.

Everything was built inside the crate the audit covered (`crates/hashed-ecdl-tensor`, Rust): a multiplier switch so `modular_multiply` uses either Theorem 3′s Fourier gadget (the default, unchanged) or a schoolbook AND/ripple gadget; the small-x factor base $x < 2^l$ (high bits constant zero, no membership gadget); a Semaev S_3 gadget; four languages: **grouplaw-hash** (Bochum Lemma 11, $m = 2$, derived last slot, the crate's own language), **grouplaw-smallx** (same group law, small-x base), **semaev-s3-smallx** ($S_3(x_1, x_2, x_R) = 0$ on abscissae only, $m = 2$) and **chained-s3-smallx** ($S_3(x_1, x_2, t) = 0$, $S_3(t, x_3, x_R) = 0$, $m = 3$, the sparse form of S_4); and a width module (Bodlaender–Koster simplicial / almost-simplicial safe reductions, MMD+ lower bounds with min-d and least-c contraction, min-degree, eager and lazy min-fill orders with seeded restarts, every width recomputed exactly from the executed order, PACE `.gr` export and `.td` verifier). The census binary `tensor-treewidth` finds one prime-order curve per b (random $p \equiv 3 \bmod 4$, BSGS point count, certified by `Curve::new`), plants a target as a sum of factor-base points, builds every (relation, multiplier) network, measures it, and at $b \le 10$ evaluates the network on every candidate of its language against direct field arithmetic.

2. Protocol

- Curves: one per $b \in \{8, 10, 12, 14, 16, 20, 24, 28, 32, 36, 40\}$, seed $20261111 \oplus (b \ll 32)$; parameters recorded per row. Slots: $m = 2$, $l = \lceil b/2 \rceil$; chained S_3: $m = 3$, $l = \lceil b/3 \rceil$. Hash rows: s_H = first prime $\ge 2p$, seeds $(3, 1)$, $(5, 2)$, $\tau = \lceil s_H(\lfloor\sqrt{r}\rfloor + 1)/r \rceil$.
- Validation ($b \le 10$, schoolbook): every small-x point, every (x_1, x_2) box, every positive tuple of the chained S_3 found by scanning t over $\mathbb{F}_p$ plus 200 random tuples, every affine point for the hashed rows; 0 mismatches required, positives counted.
- Width per network: clique bound, MMD+ on the raw graph, safe reductions, MMD+ on the core (lower bound = max of the three); min-degree, eager min-fill ($b \le 16$, and $b = 20, 24$ for the two smallest relations with one restart), lazy min-fill, 3 restarts each where marked; best upper = minimum. Fourier graphs: min-degree only (vertex arity 132–520; the clique bound dominates).
- Cost: $\log_2\left(\sum_{\text{vertices}} 2^{\text{arity}} + \sum_v 2^{|N(v)|+1}\right)$ along the best order **[exact]**.

3. Recorded rows

multiplier	relation	b	l	restarts	s (edges)	V	max arity	free bits	core vars	tw lower	min-degree	min-fill	min-fill lazy	best upper	log2 cost (best)	s/ sqrt(log2 s)	enum (m-1)b/m	validated
fourier	chained-s3-smallx	8	3	1	21,437	5,423	132	17	14,943	139	189	-	-	189	191.0	5652	5.33	-
fourier	chained-s3-smallx	12	4	1	40,688	7,200	132	24	35,841	206	341	-	-	341	343.0	10398	8.00	-
fourier	chained-s3-smallx	16	6	1	42,250	10,182	260	34	29,734	275	389	-	-	389	392.0	10778	10.67	-
fourier	chained-s3-smallx	20	7	1	78,589	12,567	260	41	69,416	402	669	-	-	669	671.0	19488	13.33	-
fourier	chained-s3-smallx	24	8	1	80,656	13,720	260	48	71,365	405	673	-	-	673	675.0	19978	16.00	-
fourier	grouplaw-hash	8	4	1	59,287	15,134	132	16	41,849	197	263	-	-	263	267.8	14889	4.00	-

fourier	grouplaw-hash	12	6	1	108,127	20,122	132	24	95,226	206	337	-	-	337	339.0	26441	6.00	-
fourier	grouplaw-hash	16	8	1	116,807	28,422	260	32	83,137	389	528	-	-	528	530.0	28469	8.00	-
fourier	grouplaw-hash	20	10	1	208,319	34,906	260	40	183,694	404	870	-	-	870	872.0	49560	10.00	-
fourier	grouplaw-hash	24	12	1	214,125	38,297	260	48	189,340	405	734	-	-	734	736.0	50884	12.00	-
fourier	grouplaw-smallx	8	4	1	29,626	7,724	132	12	20,338	139	185	-	-	185	187.0	7687	4.00	-
fourier	grouplaw-smallx	12	6	1	56,212	10,314	132	18	49,307	205	326	-	-	326	328.0	14151	6.00	-
fourier	grouplaw-smallx	16	8	1	58,386	14,514	260	24	40,446	275	405	-	-	405	407.0	14673	8.00	-
fourier	grouplaw-smallx	20	10	1	108,432	17,922	260	30	95,403	403	731	-	-	731	733.0	26513	10.00	-
fourier	grouplaw-smallx	24	12	1	111,348	19,633	260	36	98,073	405	672	-	-	672	674.0	27195	12.00	-
fourier	semaev-s3-smallx	8	4	1	10,672	2,692	132	8	7,358	139	181	-	-	181	183.0	2917	4.00	-
fourier	semaev-s3-smallx	12	6	1	20,284	3,576	132	12	17,834	205	335	-	-	335	337.0	5362	6.00	-
fourier	semaev-s3-smallx	16	8	1	21,032	5,052	260	16	14,638	275	372	-	-	372	374.0	5550	8.00	-
fourier	semaev-s3-smallx	20	10	1	39,188	6,240	260	20	34,554	401	658	-	-	658	660.0	10032	10.00	-
fourier	semaev-s3-smallx	24	12	1	40,208	6,812	260	24	35,506	403	639	-	-	639	641.0	10281	12.00	-
fourier	semaev-s3-smallx	28	14	1	41,036	7,288	260	28	37,306	405	640	-	-	640	643.0	10483	14.00	-
fourier	semaev-s3-smallx	36	18	1	74,756	11,200	520	36	67,258	788	1477	-	-	1477	1479.0	18579	18.00	-
schoolbook	chained-s3-smallx	8	3	3	14,864	8,725	5	17	13,691	24	116	78	111	78	81.6	3993	5.33	pass 13/648
schoolbook	chained-s3-smallx	10	4	3	20,386	11,856	5	22	19,042	28	154	105	138	105	107.0	5388	6.67	pass 6/4104
schoolbook	chained-s3-smallx	12	4	3	24,984	14,532	5	24	23,483	32	205	131	190	131	133.0	6537	8.00	-
schoolbook	chained-s3-smallx	14	5	3	34,314	19,681	5	29	32,608	37	198	151	222	151	153.2	8840	9.33	-
schoolbook	chained-s3-smallx	16	6	3	47,418	26,782	5	34	45,488	41	272	179	269	179	182.0	12031	10.67	-
schoolbook	chained-s3-smallx	20	7	3	75,964	42,235	5	41	73,603	48	339	-	328	328	330.1	18866	13.33	-
schoolbook	chained-s3-smallx	24	8	3	111,344	61,268	5	48	108,553	60	406	-	401	401	403.0	27194	16.00	-
schoolbook	chained-s3-smallx	28	10	3	122,794	68,348	5	58	119,761	68	579	-	562	562	564.0	29865	18.67	-
schoolbook	chained-s3-smallx	32	11	3	174,712	96,025	5	65	171,210	75	690	-	607	607	609.0	41866	21.33	-
schoolbook	chained-s3-smallx	36	12	3	212,240	116,562	5	72	208,377	84	851	-	670	670	672.0	50454	24.00	-
schoolbook	chained-s3-smallx	40	14	1	263,674	144,300	5	82	259,408	93	2303	-	861	861	863.0	62134	26.67	-
schoolbook	grouplaw-hash	8	4	3	37,619	22,416	5	16	34,402	27	130	92	162	92	95.0	9649	4.00	pass 1/232
schoolbook	grouplaw-hash	10	5	3	51,037	30,089	5	20	47,307	31	173	127	192	127	129.1	12906	5.00	pass 2/852
schoolbook	grouplaw-hash	12	6	3	61,639	36,374	5	24	57,459	36	224	153	246	153	155.0	15453	6.00	-
schoolbook	grouplaw-hash	14	7	3	83,325	48,429	5	28	78,565	40	276	171	319	171	174.0	20609	7.00	-
schoolbook	grouplaw-hash	16	8	3	114,463	65,360	5	32	109,071	48	303	205	364	205	207.0	27922	8.00	-
schoolbook	grouplaw-hash	20	10	3	184,015	103,083	5	40	177,395	56	449	-	395	395	398.0	44001	10.00	-
schoolbook	grouplaw-hash	24	12	3	272,277	150,542	5	48	264,425	67	599	-	593	593	595.0	64079	12.00	-
schoolbook	grouplaw-hash	28	14	3	295,995	165,619	5	56	287,381	75	827	-	639	639	641.0	69430	14.00	-
schoolbook	grouplaw-hash	32	16	3	412,079	227,751	5	64	402,207	84	880	-	798	798	800.0	95414	16.00	-
schoolbook	grouplaw-hash	36	18	3	521,803	286,964	5	72	510,771	95	941	-	900	900	902.0	119731	18.00	-
schoolbook	grouplaw-hash	40	20	1	624,865	343,129	5	80	612,769	104	3048	-	1157	1157	1159.0	142408	20.00	-
schoolbook	grouplaw-smallx	8	4	3	19,274	11,472	5	12	17,527	26	126	77	146	77	79.5	5109	4.00	pass 2/82
schoolbook	grouplaw-smallx	10	5	3	24,745	14,657	5	15	22,744	31	156	104	162	104	106.0	6477	5.00	pass 4/96
schoolbook	grouplaw-smallx	12	6	3	30,180	17,862	5	18	27,923	35	183	128	191	128	131.2	7823	6.00	-
schoolbook	grouplaw-smallx	14	7	3	40,919	23,824	5	21	38,346	39	268	158	247	158	160.3	10454	7.00	-
schoolbook	grouplaw-smallx	16	8	3	56,786	32,426	5	24	53,866	43	253	191	261	191	193.0	14289	8.00	-
schoolbook	grouplaw-smallx	20	10	3	91,896	51,413	5	30	88,304	54	386	-	380	380	382.0	22632	10.00	-
schoolbook	grouplaw-smallx	20	10	1	92,656	51,801	5	30	89,057	51	736	338	423	338	340.0	22811	10.00	-
schoolbook	grouplaw-smallx	24	12	3	137,300	75,726	5	36	133,032	61	484	-	485	484	486.0	33235	12.00	-

schoolbook	grouplaw-smallx	24	12	1	136,676	75,408	5	36	132,413	63	792	395	554	395	397.6	33090	12.00	-
schoolbook	grouplaw-smallx	28	14	3	146,984	82,083	5	42	142,296	70	655	-	559	559	561.0	35477	14.00	-
schoolbook	grouplaw-smallx	32	16	3	204,634	112,854	5	48	199,270	77	634	-	661	634	636.0	48719	16.00	-
schoolbook	grouplaw-smallx	36	18	3	260,236	142,726	5	54	254,233	86	807	-	752	752	754.3	61356	18.00	-
schoolbook	grouplaw-smallx	40	20	1	311,592	170,593	5	60	305,006	99	2655	-	988	988	990.0	72940	20.00	-
schoolbook	semaev-s3-smallx	8	4	3	7,104	4,169	5	8	6,500	22	97	71	99	71	73.2	1986	4.00	pass 2/256
schoolbook	semaev-s3-smallx	10	5	3	9,387	5,471	5	10	8,698	26	126	93	125	93	95.4	2584	5.00	pass 2/1024
schoolbook	semaev-s3-smallx	12	6	3	11,528	6,714	5	12	10,756	31	169	122	146	122	124.1	3138	6.00	-
schoolbook	semaev-s3-smallx	14	7	3	15,619	8,971	5	14	14,742	35	205	146	184	146	148.3	4185	7.00	-
schoolbook	semaev-s3-smallx	16	8	3	21,640	12,223	5	16	20,647	39	217	180	229	180	182.0	5702	8.00	-
schoolbook	semaev-s3-smallx	20	10	3	35,104	19,479	5	20	33,887	46	269	-	301	269	271.8	9034	10.00	-
schoolbook	semaev-s3-smallx	20	10	1	34,904	19,377	5	20	33,690	46	741	349	387	349	351.0	8985	10.00	-
schoolbook	semaev-s3-smallx	24	12	3	52,224	28,633	5	24	50,780	53	350	-	328	328	330.2	13192	12.00	-
schoolbook	semaev-s3-smallx	24	12	1	52,464	28,755	5	24	51,019	54	710	359	482	359	361.0	13250	12.00	-
schoolbook	semaev-s3-smallx	28	14	3	56,432	31,304	5	28	54,859	59	494	-	459	459	461.0	14204	14.00	-
schoolbook	semaev-s3-smallx	32	16	3	81,688	44,683	5	32	79,867	69	544	-	506	506	508.0	20222	16.00	-
schoolbook	semaev-s3-smallx	36	18	3	97,408	53,294	5	36	95,408	74	691	-	616	616	618.0	23928	18.00	-
schoolbook	semaev-s3-smallx	40	20	1	121,216	66,050	5	40	119,004	81	2068	-	879	879	881.0	29497	20.00	-

4. Slopes

Least squares of the recorded bounds against b **[derived]**. The "best-upper" slope mixes eager min-fill ($b \leq 16$) with the weaker lazy/min-degree orders beyond, so the eager-only slope is listed separately.

multiplier	relation	b range	n	lower-bound slope (per bit)	best-upper slope (per bit)	eager min-fill slope where run (per bit)	lower/(b/2) at max b	upper/(b/2) at max b	s slope (log-log)
schoolbook	grouplaw-hash	8-40	11	2.42	32.94	13.50	5.2	57.9	1.81
schoolbook	grouplaw-smallx	8-40	13	2.20	27.33	21.13	5.0	49.4	1.82
schoolbook	semaev-s3-smallx	8-40	13	1.83	22.69	20.28	4.0	44.0	1.84
schoolbook	chained-s3-smallx	8-40	11	2.16	24.31	12.40	4.7	43.0	1.84
fourier	grouplaw-hash	8-24	5	15.35	36.88	-	33.8	61.2	1.19
fourier	grouplaw-smallx	8-24	5	18.25	34.48	-	33.8	56.0	1.22
fourier	semaev-s3-smallx	8-36	7	20.72	40.36	-	43.8	82.1	1.20
fourier	chained-s3-smallx	8-24	5	18.20	32.40	-	33.8	56.1	1.22

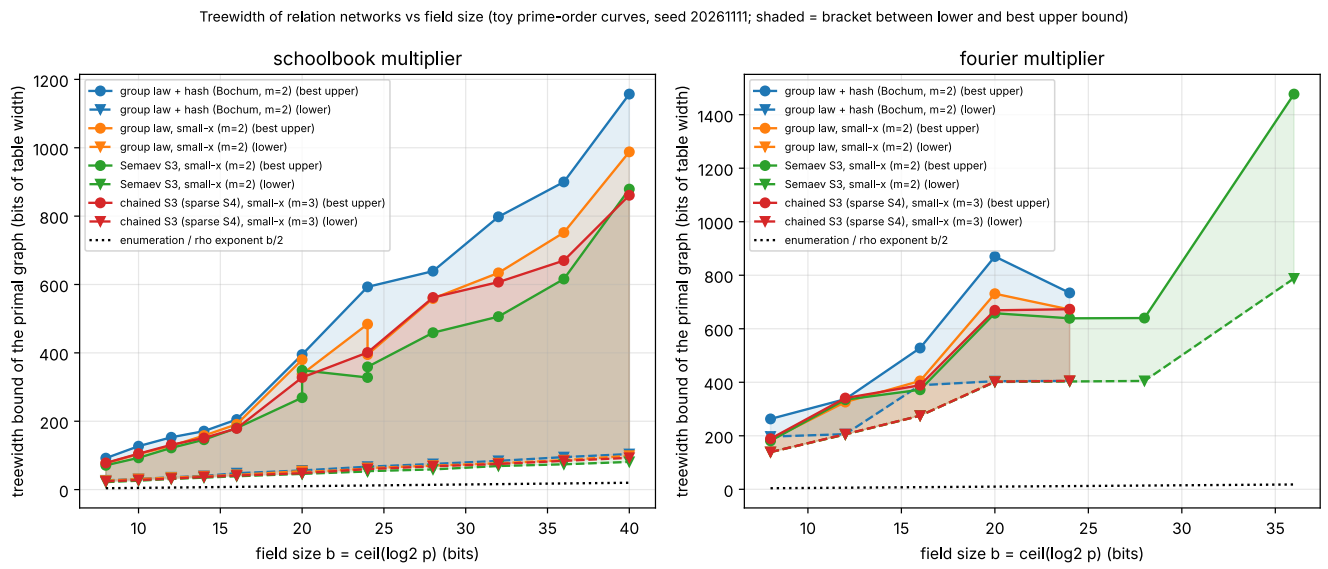


Figure 1: Treewidth bounds of the primal graph against b; shaded bands are the bracket between the lower bound **[bound]** and the best upper bound **[heuristic, exact for its order]**. Dotted: the enumeration and rho exponent $b/2$. Seed 20261111, one curve per b.

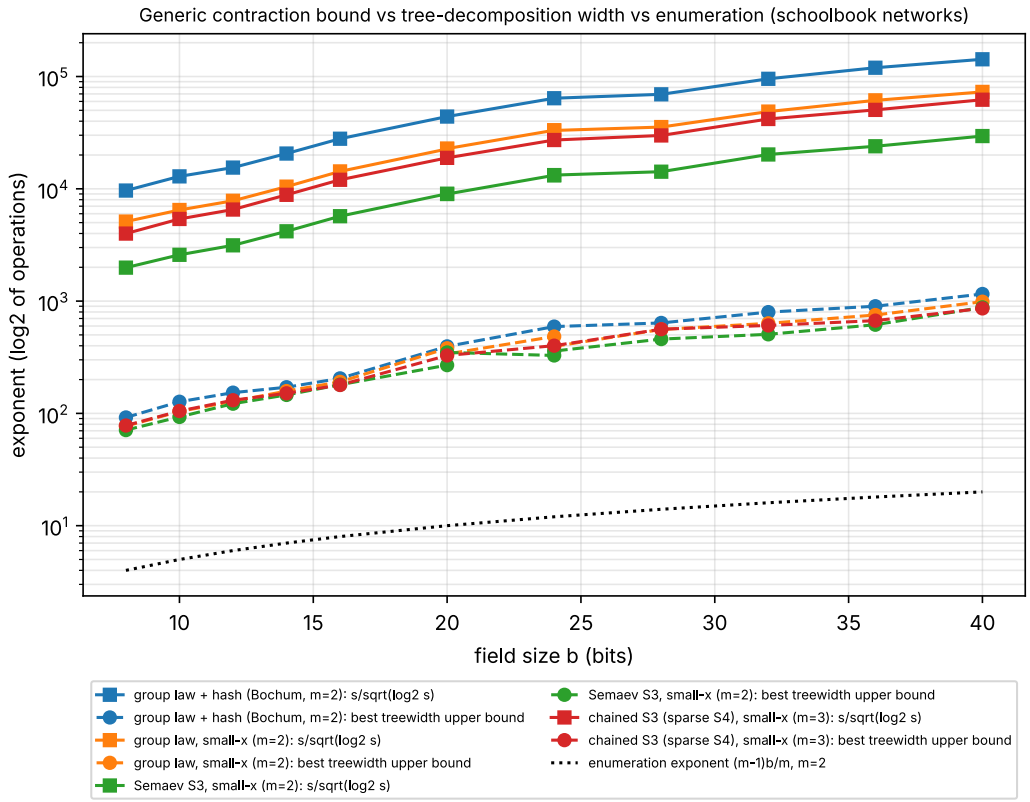


Figure 2: The three exponents on the schoolbook networks: the generic D.1 bound $s/\sqrt{\log_2 s}$ (constant 1), the best treewidth upper bound, and naive enumeration $(m-1)b/m$.

5. Reading

Bounds against enumeration. On every schoolbook row the treewidth lower bound **[bound]** is 4–5 times the enumeration exponent $b/2$ and grows at about 2 per bit against 0.5 per bit: S_3 gives $22, 39, 53, 69, 74, 81$ at $b = 8, 16, 24, 32, 36, 40$ ($b/2 = 4, 8, 12, 16, 18, 20$); the hashed group law $27, 48, 67, 84, 95, 104$. A lower bound on width binds every elimination order, so an exact contraction of these networks costs at least $2^{\approx 2b}$ table operations where enumerating one free slot costs $2^{b/2}$: no solver that treats the bit-level network as a Boolean tensor network reaches enumeration, whatever its order-finding. The best measured orders **[heuristic, exact for their order]** sit at 71–92 ($b = 8$), 180–205 ($b = 16$) and 616–900 ($b = 36$), and the $\log_2$ table work along them is the width plus 2 on every row **[exact]**.

Against the generic bound. The D.1 exponent $s/\sqrt{\log_2 s}$ (constant 1) is 30–150 times the best measured width **[derived]**: S_3 at $b = 36$: 23,928 against 616; hashed group law: 119,731 against 900. The structure a decomposition can see is real and large; the structure that remains is still a factor $2^{1.5b}$ above enumeration.

Semaev against hashed relations. At equal b (schoolbook) S_3 has 3.7–5.3 times fewer edges (97,408 against 521,803 at $b = 36$), a 15–25% lower width lower bound (74 against 95) and a 20–30% lower best upper bound (616 against 900) **[measured]**; lower-bound slope 1.86 per bit against 2.43 **[derived]**. The gain comes from dropping ordinates and curve-equation checks; the hash gadget itself costs about $2\times$ in s and at most 10% in width (hashed against small-x group law: lower 95 against 86, upper 900 against 752). The chained S_3 ($m = 3$) has the width of the group law at the same b (84 / 670 at $b = 36$) against a larger enumeration exponent $2b/3$: ratio 3.5 instead of 4.1.

The Fourier gadget. Its transform vertex is a clique of 132, 260 or 520 variables, so the width is fixed by the multiplier and not by the relation: lower 139–405 and upper 181–870 at $b = 8$–24 for all four relations, and for S_3 lower 788 / upper 1,477 at $b = 36$ once the 520-variable transform applies: 35–60 times $b/2$ (44 times at 36). The $O(b)$-edge encoding minimises s, which the generic bound charges, by maximising width, which a decomposition pays; s grows with log–log slope 1.2 against 1.8 for schoolbook **[derived]**, as the audit's gadget count predicts.

Answer. Encoding S_3/S_4 as a constraint graph does give a solver structure to exploit, but not the kind that closes the gap: every bit-level encoding of a b-bit modular multiplication has width $\Omega(b)$, measured here as at least $2b$ (lower bound) and about $20b$ (greedy upper bound), and a relation needs several of them, so the exponent is a constant multiple of b where enumeration pays $b/2$. The encoding is the bottleneck; a better contraction solver would recover two orders of magnitude of exponent and still lose. The lever that remains is algebraic, not graphical: elimination at the field level (F_4 on the S_3 system, the known index-calculus route) rather than at the bit level. The audit's closed verdict stands for the hashed encoding, and the Semaev encoding does not reopen it by this route.

Validation. Every schoolbook row at $b = 8$ and 10 passed: 0 mismatches, 1–13 positives among 82–4,104 candidates **[exact]**. Fourier rows use the crate's own tested language; the multiplier-independence test shows both gadgets accept the same set on the 17-curve.

What the runtime budget changed. The two-hour sweep budget stopped the first schoolbook sweep after 36 bits and the Fourier sweep after 24 bits. The 40-bit schoolbook rows were rerun with one restart (all four relations) and the Fourier S_3 rows at 28 and 36 bits in a second run; both are in `rows/` and in the tables above. Fourier rows for the other three relations beyond 24 bits were not run.

6. Follow-up: the field-level system (2026-11-12)

The census says every bit-level encoding has width $\geq 2b$. The remaining lever is elimination at the field level. `semaev-descent-treewidth` (`src/bin/semaev_descent_treewidth.rs`) builds the Weil-descended binary Semaev S_3 point-decomposition system exactly as the lab's refutation harness does (X_1, X_2 in the subspace V of dimension $n' = n/2$, $2n'$ Boolean variables, n quadratic equations) and measures the primal treewidth of that field-level system in a black-box encoding (one factor per equation) and an arity-3 encoding (shared AND gadgets, XOR chains), next to the refutation degree D^* the harness reaches on the same instance (`d_max` = 4). Rows `rows/field-level-8-30.json`; details in `FIELD_LEVEL.md`.

n	n'	cross monomials	bb width (exact)	split lower / upper	split log2 cost	enum one slot	D^*	Macaulay cols at D^*
8	4	16	7	12 / 19	22.3	4	2	37
12	6	36	11	16 / 39	41.4	6	2	79
15	7	49	13	22 / 54	56.4	7	4	1,471
16	8	64	15	23 / 74	76.4	8	2	137
22	11	121	21	32 / 104	106.1	11	2	254
28	14	196	27	41 / 182	184.5	14	2	407
30	15	225	29	44 / 159	162.4	15	> 4	-

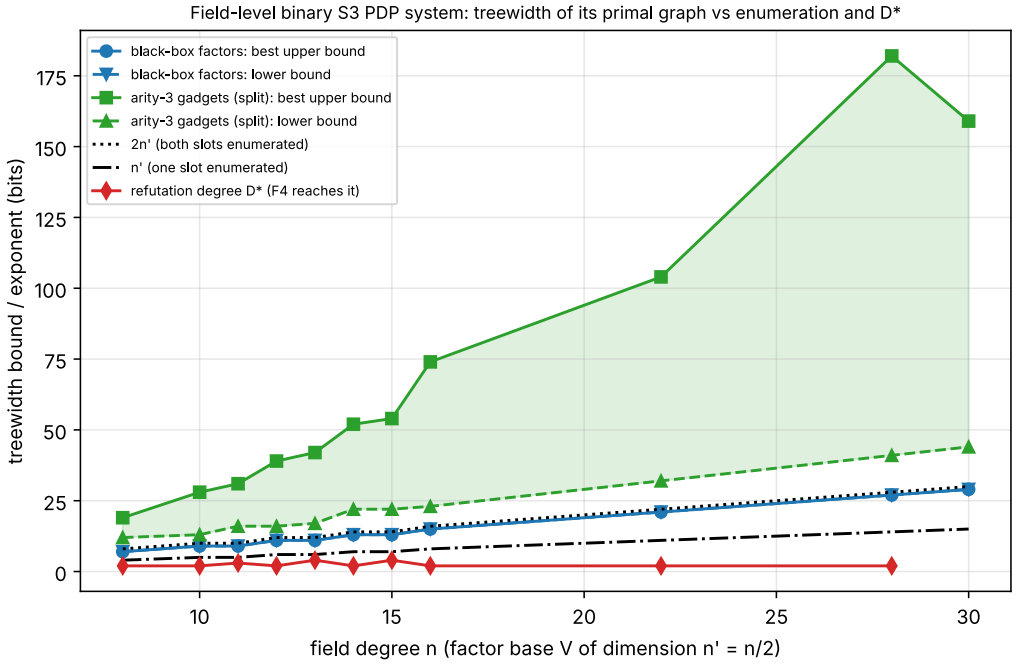


Figure 3: Field-level binary S_3 system: treewidth bounds of its primal graph (black-box and arity-3 encodings) against the one- and two-slot enumeration exponents and the measured refutation degree D^*. Seed 20261112, one instance per n.

Reading. In the black-box encoding every equation touches every variable, so the width is $2n' - 1$ exactly **[exact]**: a tree-decomposition contraction of the field-level system is enumeration of both slots, $2^{2n'}$, twice the exponent of the naive one-slot search. The arity-3 form is wider still (best upper 19 $\longrightarrow$ 182 over $n = 8 \rightarrow 30$). The refutation harness derives the contradiction on the same instances at Macaulay degree 2 (even n) or 3–4 (odd n) with 37–1,471 columns **[measured]**: polynomial in n'. The structure the field-level route exploits is algebraic degree, which a Boolean tensor network cannot represent; it is the lab's existing index-calculus program (F4 on $\mathbb{F}_2$-linear bases), and the lab's own measurements say where it stops being cheap (solving degree 7 at $m = 3$; `ml-cryptanalysis` stage 5). Over prime fields the small-x base has no low-degree membership polynomial, and the measured algebraic alternative (PKM Dickson towers) costs $2^{38.8}$ multiply-adds per attempt at $m = 2$, $N = 20$ against 2^{20} enumeration. Neither the bit-level nor the field-level graph carries the lever; the contraction-solver framing adds nothing to F4.

7. Limits

- One curve per b; s moves by a few percent with the Hamming weights of p and the planted constants (the schoolbook rows fold constant bits), so s is not monotone to the last digit.
- Upper bounds are greedy orders. The eager min-fill series is the trustworthy one; beyond $b = 16$ only lazy min-fill and min-degree ran, and their widths are noticeably worse (and noisier across tie-breaking seeds) than eager min-fill would give. The `.gr` export is there so a stronger external heuristic (FlowCutter, Tamaki) can tighten them; its output is checked by the crate's `.td` verifier, not trusted.
- Lower bounds are MMD+ on the reduced core: valid, weak. The true treewidth is somewhere in the shaded band.
- Treewidth of the primal graph is the exact-contraction cost model for **Boolean tables**; it does not credit algebraic structure inside a tensor (an F4-style Gröbner step on the S_3 system is a different algorithm and is not measured here).
- No pcc claim, no benchmark record (no timing claim), no leaderboard row.